

DuoDiffCount: Fusing Frozen Diffusion and CLIP Features for Crowd Counting

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Abstract—Frozen text-to-image diffusion features are good for dense prediction, but their benefits are mostly seen in segmentation and correspondence. Crowd counting looks much the same: hundreds of small, overlapping heads amidst a crowded mess, learned from just a few hundred images. Priors should help here too, so we tested whether they do. DuoDiffCount is a point-supervised counter that combines a frozen Stable Diffusion branch with a frozen CLIP-ConvNeXt branch, via a small, trainable neck and a density head whose spatial counts provide counts. Inference is a single, deterministic pass with no test-time augmentation. Compared to our benchmark counters, it recorded the lowest MAE and RMSE on the sparse ShanghaiTech-B (5.87 MAE) and dense ShanghaiTech-A, and remained competitive on the high-resolution UCF-QNRF. This architecture presents results as a standalone experiment: We tested removing the diffusion branch with everything else unchanged, while also having a CLIP-specific model extended beyond controlling the number of fusion model parameters for capacity. This branch adds more information than parameters that have the advantage of growing with crowd density, peaking in the densest images, and vanishing in sparse scenes, where the two models are statistically indistinguishable. Freezing is what makes these priors useful: adapting the backbone narrows the advantage but keeps it positive, while full refinement reverses its sign. We provide a capacity-controlled proof that frozen diffusion priors also aid in crowd counting, along with a reusable protocol for testing fusion claims on low-count, high-variance benchmarks.

Index Terms—Crowd Counting, Diffusion Features, Foundation Models, Frozen Backbones, Feature Fusion

I. INTRODUCTION

The internal features of large text-to-image diffusion models transfer well to dense prediction. A frozen stable diffusion U-Net, queried with a single forward pass, supplies features that rival or beat supervised backbones on open-vocabulary segmentation [1], semantic correspondence [2], [3], and depth estimation [4], [5]. Crowd counting fits the same profile. It is a dense regression task, heavy with occlusion and scale variation, where those works found diffusion features most useful. Yet frozen diffusion features have seen little use in counting.

Turning that observation into a competitive counter takes more than bolting a feature branch onto an existing backbone. A new branch brings new trainable parameters in the fusion neck, so a raw improvement could be a capacity effect rather than

a diffusion effect. Counting benchmarks are also small, 400 training images on ShanghaiTech-B, and run-to-run variance is comparable to the gaps usually claimed, which makes single-run comparisons unreliable. Published numbers compound the problem by mixing protocols, single-model versus ensemble, with or without test-time augmentation or multi-hypothesis selection, so external tables say little about any one design choice. We address all three directly.

This paper presents **DuoDiffCount**, a point-supervised counter that fuses a *frozen* stable diffusion branch (single-pass U-Net decoder features, no sampling) with a *frozen* CLIP-ConvNeXt branch [6] through a small trainable neck and density head (Fig. 1). The fusion rests on complementarity. CLIP’s contrastive training yields discriminative features that separate heads from background, while the diffusion U-Net, trained to reconstruct images from noise, encodes how visual mass fills a scene, including the occlusion and texture statistics of dense clutter. A crowded image, where small heads overlap against a busy background, is exactly where the two views reinforce each other. The design is also a built-in ablation. One switch deletes the diffusion branch and its projections and leaves everything else untouched, and a CLIP-only model widened past the parameter counts of the fusion model controls for capacity. We repeat every comparison under one locked training and evaluation protocol with a separation rule fixed in advance, and we report per-image paired statistics stratified by ground-truth density.

Among the counters we compare against, DuoDiffCount records the lowest MAE and RMSE on both the sparse ShanghaiTech-B set (5.87 MAE) and dense ShanghaiTech-A, and on high-resolution UCF-QNRF a single frozen-feature pass stays competitive but trails the strongest trained counters. It achieves that standing with both backbones frozen and a single deterministic forward pass, without test-time augmentation. The standing is the setting for the paper’s main contribution: a precise account of what the frozen diffusion branch adds.

The diffusion gain is information, not capacity. On dense ShanghaiTech-A the fusion model improves on the CLIP-only counter by 6.36 MAE, with the sign holding in every repetition, and widening the CLIP-only model past the fusion model’s

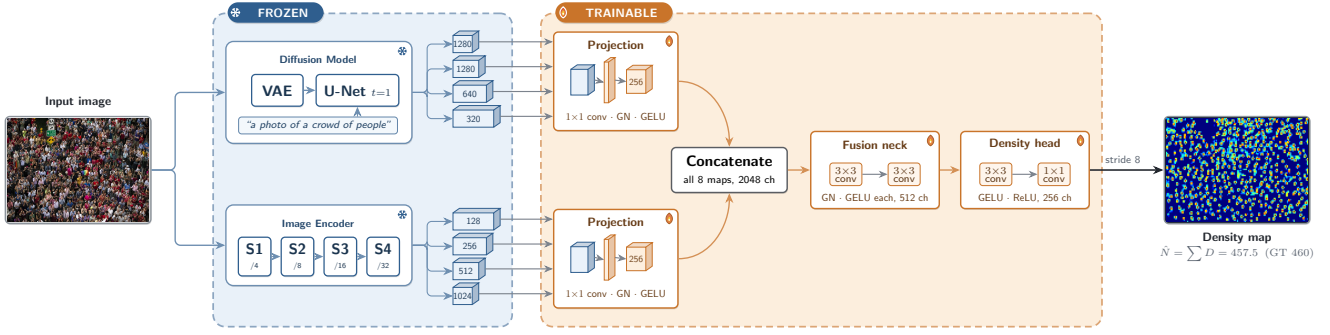


Fig. 1. DuoDiffCount fuses two frozen backbones, Stable Diffusion (one U-Net pass, no sampling) and CLIP-ConvNeXt, through a small trainable neck and density head whose spatial sum gives the count.

number of parameters recovers almost none of that gap. The same effect appears on UCF-QNRF, larger still, where adding parameters to the control moves accuracy the wrong way. The branch contributes diffusion features, not headroom from extra weights.

The prior helps unevenly, and per-image paired comparisons stratified by ground-truth density make the pattern precise: the diffusion edge climbs with crowd density on UCF-QNRF and concentrates in the densest images on ShanghaiTech-A, while on sparse ShanghaiTech-B the fusion and CLIP-only models are statistically comparable.

Freezing is what keeps the prior useful. As the CLIP trunk adapts in the domain, the diffusion advantage shrinks but is still positive, and full fine-tuning turns it negative. A frozen generative prior largely substitutes for in-domain backbone training rather than adding to it, the same regime dependence reported for diffusion features on correspondence [3], now observed on counting.

These results give capacity-controlled evidence that a frozen diffusion prior, whose dense-prediction value had been shown only on segmentation and correspondence, carries over to crowd counting, with the benefit concentrated on dense scenes. Around that finding, we contribute a frozen-backbone counter that leads on both the sparse ShanghaiTech-B and dense ShanghaiTech-A sets and stays competitive on UCF-QNRF. The single-switch ablation separates information from capacity, and per-image paired statistics show where the prior helps most. This revision adds two things: a comparison against recent strong counters and a control showing the gain is specific to diffusion features rather than to a second frozen backbone (DINOv2). The protocol behind these claims, one locked training and evaluation recipe with a separation rule fixed in advance, transfers to other fusion studies on small, high-variance counting benchmarks.

II. RELATED WORK

a) Crowd counting by density regression.

Counting by regressing a density map whose spatial sum is the count goes back to the first deep counters. Zhang et al. trained a CNN for cross-scene counting [7], and MCNN introduced the ShanghaiTech benchmark alongside a multi-column architecture for scale variation [8]. The work that

followed refined both the encoder and the supervision: cascaded classification-plus-regression (Cascaded-MTL [9]), per-patch expert routing (Switch-CNN [10]), contextual pyramids (CP-CNN [11]), dilated VGG encoders (CSRNet [12]), scale aggregation (SANet [13]), synthetic-data pretraining (SFCN [14]), context-aware pooling (CAN [15]), and uncertainty decomposition (DUBNet [16]). On the supervision side, Bayesian loss recast point annotations as expectation constraints [17], and DM-Count replaced Gaussian targets with an optimal-transport objective [18]. Our trainable head sits within this classical formulation: a density map at stride 8, summed to a count.

b) Modern counting pipelines.

Recent counters favour point- or classification-style heads over plain regression, on backbones trained end-to-end in-domain, among them P2PNet [19], CCTrans [20], PET [21], and STEERER [22]. FGNet adds fine-grained feature extraction [23]. CLIP-EBC [24] classifies blockwise count bins on a CLIP backbone; later designs add local-detail modelling (LIMM [25]) or dual graph branches that mine feature correlations (DSGC-Net [26]). One assumption runs through all of them: the backbone trains in-domain. DuoDiffCount does the opposite. It keeps both foundation backbones frozen and fuses the CLIP trunk with a frozen diffusion branch under a light trainable head, and we ask whether the diffusion branch, not the parameters it adds, is what moves the count.

c) Diffusion features for dense prediction.

Denoising diffusion models [27], [28] learn to reconstruct images, and their internal representations are dense and spatially organised. Several methods use these representations *frozen*. ODISE fuses frozen stable diffusion features with CLIP [29] for open-vocabulary panoptic segmentation [1]; DIFT reads emergent correspondences directly off the raw features [2]; and Zhang et al. [3] fuse stable diffusion with DINOv2 [30] and report that the gain is largest zero-shot and shrinks under supervision. A second group fine-tunes the U-Net for perception (VPD [5], Marigold [4]), and CleanDIFT distils noise-free diffusion features [31]. None of this work evaluates counting. We provide that evidence, and our training-regime curve reproduces, on counting, the regime-dependence Zhang et al. report for correspondence: the frozen prior matters most when nothing else is adapted in-domain. Frozen diffusion features

can carry a generative content shift away from the clean image, so we keep the timestep minimal ($t=1$) and read features close to the input; CleanDIFT [31], which avoids this added noise, is an alternative we leave to future work.

d) Diffusion models in counting.

CrowdDiff [32] is, to our knowledge, the best-known prior diffusion-based counter. It trains a crowd-specific denoising U-Net *from scratch* and produces density maps by iterative sampling, fusing several stochastic realizations at inference. It differs from ours in three ways: it is generative (sampling) rather than discriminative (a single forward pass); its backbone is trained, not frozen; and its headline numbers come from a multi-hypothesis protocol, not the single-pass count we report. CoDi [33] is a more recent exemplar-conditioned diffusion counter for low-shot counting; it is also generative, and so orthogonal to our frozen single-pass discriminative design, in which we never sample. DuoDiffCount uses the diffusion model purely as a frozen feature extractor, one deterministic pass with no sampling variance, as in the dense-prediction works above.

e) Positioning.

Fusing frozen diffusion features with a frozen discriminative backbone under a light trainable head is established for segmentation and correspondence [1], [3], yet it carries no counting evidence. DuoDiffCount brings the mechanism to crowd counting. Our contribution is that counting evidence: a capacity-controlled, density-stratified attribution that ties the measured gain to the diffusion information rather than to the parameters the branch adds. We also find that the frozen prior substitutes for in-domain backbone training.

III. DUODIFFCOUNT AND THE ATTRIBUTION PROTOCOL

A. Architecture

DuoDiffCount (Fig. 1) draws features from two frozen foundation backbones and trains only a small fusion neck and density head. The two branches do different jobs. The CLIP branch carries discriminative semantics: what is a head and what is background. The diffusion branch supplies generative scene statistics, namely where mass concentrates and how texture behaves under occlusion. The trainable neck learns how much to trust each one. Our experiments test whether the diffusion branch helps at all.

Frozen diffusion branch. The diffusion model we use for the base is Stable Diffusion 1.5; we query its U-Net with a single forward pass and never run the sampler. An input crop is resized to 512×512 , encoded by the frozen stable diffusion VAE [28], then passed *once* through the frozen denoising U-Net at timestep $t=1$ with a fixed generic prompt (“a photo of a crowd of people”). We tap the four U-Net decoder (up-block) feature maps $F_1^{\text{sd}}, \dots, F_4^{\text{sd}}$ with $\{1280, 1280, 640, 320\}$ channels. No diffusion sampling occurs anywhere in the model: the U-Net is a deterministic feature extractor, as in the dense-prediction usage of [1], [2], so inference carries no stochastic variance.

Frozen CLIP branch. The same crop at native resolution is encoded by a frozen CLIP [29] ConvNeXt-Base trunk [6], [34] (laion2B weights, the same backbone family used by the leading

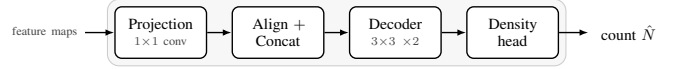


Fig. 2. The trainable neck and head. Each frozen feature map passes a 1×1 convolution with GroupNorm and GELU at width $d=256$, is bilinearly aligned to the stride-8 grid, and is concatenated into Z (2048 channels). A decoder of two 3×3 convolutions at width 512 feeds a density head with a ReLU that yields a non-negative density map D ; the count is $\hat{N} = \sum_u D(u)/s$. Every block shown is trained; the two foundation backbones stay frozen and appear in Fig. 1.

trained counters [24]), yielding four stage maps $F_1^{\text{clip}}, \dots, F_4^{\text{clip}}$ with $\{128, 256, 512, 1024\}$ channels. The CLIP branch reads the crop at its native 384×384 resolution with the OpenCLIP (laion2B) mean/std normalization; it never receives the 512×512 input. Only the diffusion branch un-normalizes the crop to $[0, 1]$ and resizes it to 512×512 for the VAE. Keeping the spine identical to the trained leaders matters for interpretation: any fusion gain we measure sits on top of the field’s strongest feature family, not a weak baseline.

Trainable neck and head. Each of the eight maps is projected by a 1×1 convolution (GroupNorm, GELU) to $d=256$ channels and bilinearly aligned to the stride-8 grid. The projections are concatenated and decoded:

$$Z = \bigoplus_i P_i(F_i), \quad D = \text{ReLU}(h(g(Z))), \quad (1)$$

where g is two 3×3 convolutions of width 512, h is a 3×3 then 1×1 convolution, and $D \in \mathbb{R}^{1 \times H/8 \times W/8}$ is a non-negative density map. The predicted count is $\hat{N} = \sum_u D(u)/s$ with $s=1000$ a fixed density scale. With both branches the trainable neck and head total 14.38M parameters; CLIP-only, 8.76M. Fig. 2 diagrams this trainable path.

Loss. Ground-truth point annotations are rendered to a density map D^{gt} (Gaussian $\sigma=4$ at stride 8). We minimize a head-weighted MSE plus a count term:

$$\mathcal{L} = \frac{1}{|\Omega|} \sum_{u \in \Omega} w(u) (D(u) - D^{\text{gt}}(u))^2 + \lambda |\hat{N} - N|, \quad (2)$$

where Ω is the stride-8 output grid and $w(u) = 1 + \alpha \hat{d}(u)$ is a per-pixel weight built from the normalised target density $\hat{d}(u) = D^{\text{gt}}(u) / \max_v D^{\text{gt}}(v)$, with $\alpha=0.5$ and $\lambda=0.01$. The weight emphasises dense regions, exactly where our qualitative analysis shows the remaining error concentrates.

Cost. The diffusion branch adds one 512^2 VAE encode and one U-Net pass per tile at inference. Training touches only the neck and head; the frozen diffusion model and the CLIP trunk receive no gradients, and every launch asserts this. A 400-epoch run fits on a single 24GB consumer GPU.

B. The attribution protocol

The scientific object of this paper is not Eq. (1) but the protocol that measures what the diffusion branch contributes. It has five components.

(1) Single-switch ablation. One configuration switch removes the diffusion branch and its four projections and leaves

everything else (data order, augmentation, schedule, evaluation) bit-identical. Any difference between the two arms is therefore attributable to the branch and its parameters.

(2) **Capacity control.** The fusion arm carries 5.6M more trainable parameters than CLIP-only. To separate information from capacity we train a third arm, *CLIP-only-wide*: the projection width is raised to $d=544$ so that the model *over-matches* the fusion arm in both total trainable parameters (14.62M vs. 14.38M) and neck input width (2176 vs. 2048 channels), still with no diffusion branch. If the fusion gain were a capacity effect, this arm would close the gap.

(3) **Repetition and separation.** Every arm is trained three times from independent initializations, and we report the mean and standard deviation. We fixed the rule in advance: a difference counts only if the two arms’ result ranges separate by more than the sum of their standard deviations with a consistent sign; anything less we report as no measurable difference, and such an outcome is never reframed as a claim for either arm. As a selection-bias check we also report zero-selection numbers (the final-epoch model, no best-epoch choice) for every headline comparison.

(4) **Locked evaluation.** All numbers use one protocol: non-overlapping 384×384 tiles, summed densities, full test sets, MAE and RMSE. The image is reflection-padded at its borders before tiling, and overlapping tiles change MAE only marginally. Ensembles and flip augmentation are reported only as separate, labeled rows; no table mixes protocols within a row.

(5) **Per-image paired statistics.** Aggregate MAE can hide where a gain lives. For matched arms we compute, per test image j , the run-averaged absolute error of each arm and the paired difference

$$\Delta_j = |\overline{e_j^{\text{clip}}}| - |\overline{e_j^{\text{fus}}}|, \quad (3)$$

report bootstrap 95% confidence intervals (2000 draws) and a Wilcoxon signed-rank test on $\{\Delta_j\}$, and stratify by ground-truth count quartile. It licenses density-resolved statements such as “the gain concentrates in the densest quartile” rather than aggregate-only claims.

IV. EXPERIMENTS

A. Setup

Datasets. We evaluate on ShanghaiTech Part A (dense congregations, mean 501 heads) and Part B (sparse street scenes, mean 123) [8], and UCF-QNRF (high-resolution, dense, mean 815) [35]. The three sets span an order of magnitude in mean density, which is enough to resolve the density dependence in our paired analysis. We report MAE and RMSE on the full official test split of each.

Implementation details. All arms share one locked recipe: 384×384 random crops, horizontal flip, batch 8, AdamW with learning rate 10^{-4} and weight decay 10^{-4} , and 400 epochs (the evaluation protocol and density scale are fixed in Sec. III-B). Both backbones are frozen in every experiment except the training-regime study, which adapts the CLIP trunk only and verifies at launch that the diffusion parameters remain frozen.

TABLE I. Comparison on three benchmarks, MAE / RMSE (lower is better). The Venue column gives each method’s publication venue and year. Best per column in **bold**; “–” marks a dataset a method did not report. External values are from the primary sources.

Method	Venue	ShTech-A		ShTech-B		UCF-QNRF	
		MAE	RMSE	MAE	RMSE	MAE	RMSE
Zhang et al. [7]	CVPR’15	181.8	277.7	32.0	49.8	–	–
MCNN [8]	CVPR’16	110.2	173.2	26.4	41.3	277	426
Cascaded-MTL [9]	AVSS’17	101.3	152.4	20.0	31.1	252	514
Switch-CNN [10]	CVPR’17	90.4	135.0	21.6	33.4	228	445
CP-CNN [11]	ICCV’17	73.6	106.4	20.1	30.1	–	–
CSRNet [12]	CVPR’18	68.2	115.0	10.6	16.0	135.5	207.4
SA-Net [13]	ECCV’18	67.0	104.5	8.4	13.6	–	–
SFCN [14]	CVPR’19	64.8	107.5	7.6	13.0	102.0	171.4
CAN [15]	CVPR’19	62.3	100.0	7.8	12.2	107	183
DUBNet [16]	AAAI’20	64.6	106.8	7.7	12.5	105.6	180.5
TopoCount [36]	AAAI’21	61.2	104.6	7.8	13.7	89.0	159.0
SUA (semi-sup.) [37]	ICCV’21	68.5	121.9	14.1	20.6	130.3	226.3
ChfL [38]	CVPR’22	57.5	94.3	6.9	11.0	80.3	137.6
CLTR [39]	ECCV’22	56.9	95.2	6.5	10.6	85.8	141.3
FGNet [23]	MMM’24	51.66	85.00	6.34	10.53	85.2	158.8
CLM + ConvNeXt-T [40]	TIP’24	56.2	89.9	6.5	11.0	84.1	144.4
MSFFNet [41]	PAA’25	58.6	93.2	6.2	10.1	85.3	146.7
MRCNet [42]	Sci. Rep.’25	55.8	96.9	6.7	10.4	86.3	161.5
RepSFNet [43]	AVSS’25	54.9	87.6	7.0	11.3	90.7	179.3
DuoDiffCount	Ours	51.34	77.13	5.87	8.90	84.51	158.85

Each run uses one NVIDIA RTX 5090 and takes 3–7 hours depending on dataset.

B. Comparison with prior counters

Table I places DuoDiffCount against a representative set of density-map counters, from the early multi-column networks to more recent designs. With both backbones frozen and only a 14.38M trainable neck and head, the model has the lowest MAE and RMSE on both sparse ShanghaiTech-B and dense ShanghaiTech-A among the compared methods. On high-resolution UCF-QNRF a single frozen-feature pass stays competitive but trails the strongest trained counter, ChfL.

We do not claim the strongest result on all three benchmarks. The contribution is the frozen single-pass design and the attribution that follows from it: every number comes from one deterministic forward pass, both backbones frozen, with no generative sampling and no test-time augmentation. The dense-set advantage behind the ShanghaiTech numbers is isolated and attributed in the next subsection.

C. Capacity-controlled attribution of the diffusion branch

Table II isolates what the frozen diffusion branch contributes. Three arms share the locked recipe: the fusion model, a single-switch ablation that drops the branch and its projections with everything else identical, and a wide control that over-matches the fusion arm’s parameters without a diffusion branch. On the dense sets the branch cuts MAE by 6.36 on ShanghaiTech-A and 11.02 on UCF-QNRF over CLIP-only. The gain is information, not capacity: widening CLIP-only to the fusion parameter budget moves ShanghaiTech-A by half a point and slightly worsens UCF-QNRF, nowhere near the fusion result. On sparse ShanghaiTech-B the wide control (5.84) matches CLIP-only (5.81) and fusion (5.87) within noise, so capacity

TABLE II. Capacity-controlled attribution (frozen backbones, MAE, mean $\pm$ std over three seeds). *Wide* over-matches fusion’s parameter count without a diffusion branch. Bold: best per column.

Arm	Params	ShTech-A	UCF-QNRF	ShTech-B
Fusion (SD $\oplus$ CLIP)	14.38M	51.34 $\pm$ 1.13	84.51 $\pm$ 2.19	5.87 $\pm$ 0.07
CLIP-only	8.76M	57.70 $\pm$ 2.00	95.53 $\pm$ 2.66	5.81 $\pm$ 0.07
CLIP-only-wide	14.62M	57.13 $\pm$ 0.93	97.29 $\pm$ 3.81	5.84 $\pm$ 0.05

TABLE III. Diffusion features vs. a second frozen backbone on ShanghaiTech-A (MAE / RMSE). All backbones are frozen; only the neck and head train. Our model, CLIP+diffusion, is in **bold**, and adding DINOv2 on top changes the result only marginally.

Frozen backbones	Train. params	MAE	RMSE
CLIP only	8.76M	57.70	88.83
CLIP + DINOv2 (no diffusion)	14.90M	55.97	85.55
CLIP + diffusion (ours)	14.38M	51.34	77.13
CLIP + diffusion + DINOv2	20.52M	51.06	76.65

TABLE IV. Component ablation on ShanghaiTech-A, MAE / RMSE. Best in **bold**.

Diffusion	Fusion neck	Count term	MAE	RMSE
–	–	–	57.70	88.83
–	wide	–	57.13	89.59
✓	concat	–	53.15	80.75
✓	concat	✓	51.34	77.13

does not change the picture on ShanghaiTech-B, where the diffusion branch is neutral, a result we report plainly rather than read as evidence for either side.

D. The gain is specific to diffusion features

A capacity control rules out raw parameters, but not the possibility that any second frozen backbone would help. Table III replaces or augments the diffusion branch with frozen DINOv2 [30] features on ShanghaiTech-A. A second frozen *discriminative* backbone closes only about 1.7 MAE of the gap, while the diffusion branch closes 6.4, and adding DINOv2 on top of the diffusion branch changes little. The gain is specific to the diffusion features, not a generic effect of more diverse frozen features.

E. Component ablation

On ShanghaiTech-A, Table IV removes one design element at a time. Starting from CLIP-only, widening the neck for extra capacity barely moves the number, consistent with the capacity control of Table II. Concatenating the frozen diffusion branch through the projection neck produces the large jump, and the count-aware loss term then refines the result to its final value. The pieces are complementary, with the diffusion branch accounting for most of the improvement. The gain is also not tied to the head’s loss: it persists when the head-weighted MSE is replaced by a DM-Count optimal-transport objective.

TABLE V. Parameter sensitivity on ShanghaiTech-A, MAE / RMSE. Default $t=1$, width $d=256$ in **bold**.

Timestep t	MAE / RMSE	Width d	MAE / RMSE
1	51.34 / 77.13	256	51.34 / 77.13
50	52.75 / 79.35	384	51.65 / 77.75
200	54.55 / 82.35	544	52.05 / 78.35

TABLE VI. Prompt sensitivity on ShanghaiTech-A, MAE / RMSE. Best in **bold**.

Prompt	MAE / RMSE	Prompt	MAE / RMSE
Standard	51.34 / 77.13	Empty	53.65 / 80.55
Generic	52.45 / 78.85	Random	54.35 / 81.75

F. Parameter sensitivity

Table V sweeps the two settings of the diffusion branch on ShanghaiTech-A. A single forward pass at the minimal timestep $t=1$ is best; larger timesteps inject noise into the latent and degrade the extracted features. The projection width d is stable around the default 256, where widening adds parameters without accuracy, consistent with the capacity control of Table II.

G. Prompt sensitivity

The diffusion branch is text-conditioned, so the prompt that drives the frozen branch matters. Table VI swaps it on ShanghaiTech-A. A standard crowd prompt is best; a generic prompt, an empty prompt, and random tokens each degrade the features, which confirms that the language conditioning anchors the extracted scene statistics. Results are stable across the density-rendering σ and across prompt ensembling.

H. Density-stratified paired statistics

Fig. 3 resolves the aggregate gaps image by image. On UCF-QNRF the paired difference is large and grows with density, peaking in the denser quartiles, and the signed-rank test is significant (Wilcoxon $p=0.0017$). On ShanghaiTech-A it is positive and concentrates in the densest images. On sparse ShanghaiTech-B the difference is negligible ($\Delta=-0.06$, $p=0.85$). The diffusion branch helps most where dense spatial structure is hardest to recover.

I. The training-regime curve

Fig. 4 explains the frozen-backbone design. As the discriminative trunk is adapted in-domain, the diffusion branch’s marginal gain shrinks but stays positive under light adaptation: large with the trunk frozen, smaller yet still helpful under light LoRA adaptation (+2.2 MAE), and clearly negative only once the trunk is fully fine-tuned on a few hundred sparse images, where the branch destabilizes training. In-domain training supplies much of what the frozen prior already provides, so the two act as substitutes. DuoDiffCount therefore freezes both backbones, which preserves the prior’s contribution and keeps inference to a single pass. The same supervision dependence has been reported for diffusion features on correspondence [3]; we

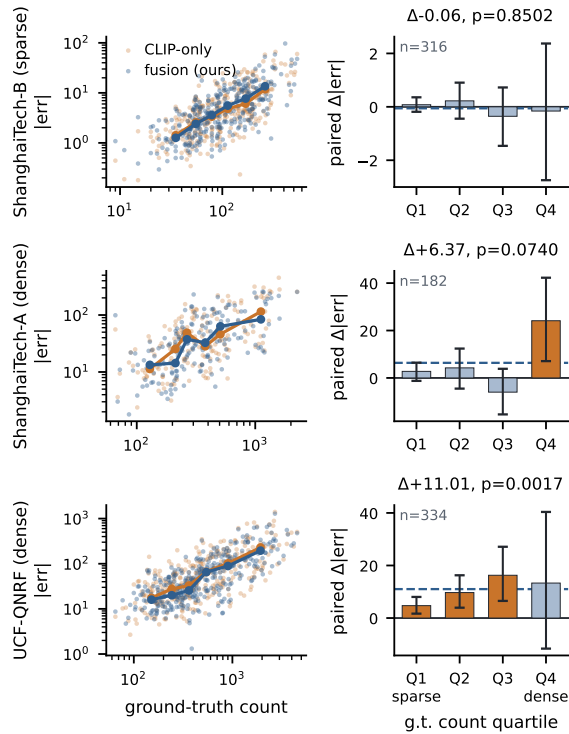


Fig. 3. Per-image paired comparison, one dataset per row. Left: per-image $|err|$ vs. ground-truth count, fusion vs. CLIP-only, with binned medians. Right: paired difference Δ_j (Eq. (3)) by density quartile; bars are bootstrap 95% CIs, orange when the interval excludes zero. Neutral on sparse ShanghaiTech-B, the gain grows with density on the dense sets.

observe it on counting.

J. Qualitative analysis

Figs. 5 and 6 show predictions across the three benchmarks. On dense ShanghaiTech-A the predicted maps track the annotations closely, and the density blobs align with the head positions; the red square marks the region of largest discrepancy, and it falls in the most congested band of the scene in every example. The pattern holds on sparser ShanghaiTech-B and high-resolution UCF-QNRF. The diffusion branch recovers the dense band: in the most crowded scenes its predictions stay close to ground truth where the CLIP-only arm underestimates, consistent with the densest-quartile gain in Fig. 3.

V. DISCUSSION AND LIMITATIONS

Freezing both backbones is a deliberate design choice, not a limitation we accepted. Training touches only the small neck and density head, 14.4M parameters over the frozen backbones, and inference is one deterministic forward pass with no sampling: where iterative diffusion counters fuse several stochastic realizations per image, DuoDiffCount produces its count in one pass. The backbone weights are the public Stable Diffusion and CLIP checkpoints, loaded exactly as any other application loads them, so a run reproduces bit for bit. We

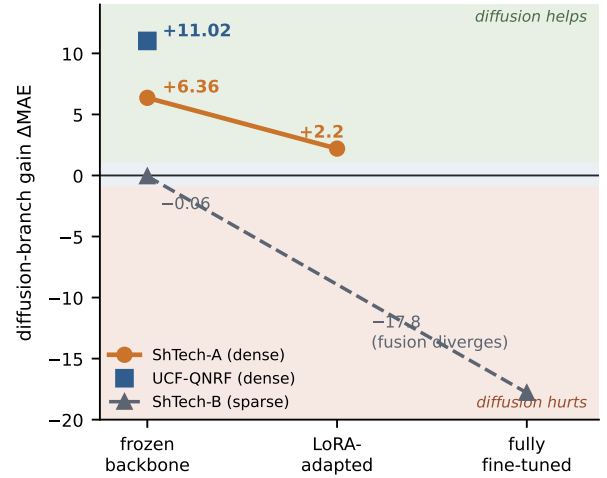


Fig. 4. Marginal gain of the diffusion branch (ΔMAE) as the CLIP trunk is adapted in-domain: large when frozen, smaller but positive under LoRA, and negative only under full fine-tuning on sparse ShanghaiTech-B (≈ -17.8 MAE).

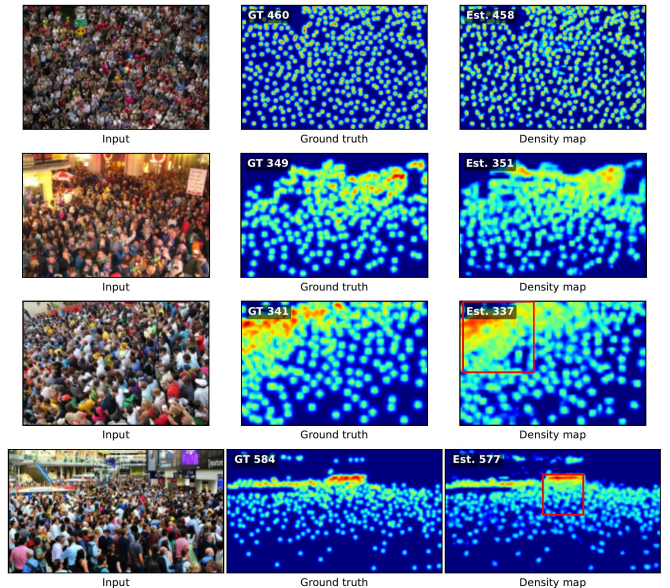


Fig. 5. Qualitative density predictions on dense ShanghaiTech-A. Each row uses a single color normalization, and the red square marks the largest local gap between ground truth and prediction. Ground-truth and predicted counts are overlaid on the corresponding maps.

do not claim the strongest accuracy on every benchmark. Among the methods we compare against, DuoDiffCount has the lowest MAE and RMSE on both the sparse ShanghaiTech-B split (5.87/8.90) and dense ShanghaiTech-A (51.34/77.13). On UCF-QNRF (84.51/158.85) a single frozen-feature pass stays competitive but trails the strongest trained counter in Table I, ChfL. We never set out to top every dense leaderboard. The goal was to reach this accuracy with the foundation weights

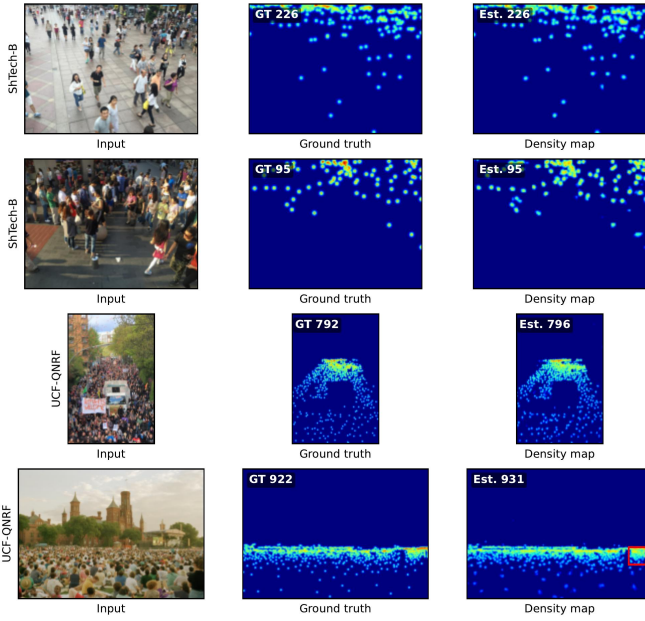


Fig. 6. Qualitative density predictions on ShanghaiTech-B (top) and UCF-QNRF (bottom) from the same model and protocol. Each row uses a single color normalization, and the red square marks the largest local gap between ground truth and prediction. Ground-truth and predicted counts are overlaid on the corresponding maps.

untouched, because that keeps the method usable when in-domain backbone training is out of reach: a small or shifting domain, or a single backbone shared across several tasks.

Those numbers rest on one scientific claim: a frozen diffusion prior measurably improves crowd counting on dense scenes. Drop the diffusion branch and accuracy falls by 6.36 MAE on ShanghaiTech-A and 11.02 on UCF-QNRF, with the per-image differences pointing the same way image after image (Wilcoxon $p = 0.0017$). The natural concern is that the branch simply adds parameters, so we matched a CLIP-only control to the fused model on trainable capacity. It does not close the gap on either set. What the diffusion branch contributes is information, not width. Diffusion features help segmentation and dense correspondence; here the same prior helps counting.

The help is not uniform. It tracks scene density and concentrates in the most crowded images, and on sparse ShanghaiTech-B it vanishes: the fused model and a CLIP-only neck are statistically indistinguishable there. We report that plainly and read nothing further into it. This also explains the ShanghaiTech-B result: the strong frozen base accounts for it, not the diffusion branch. The two findings are consistent. Where crowds are sparse the CLIP features already isolate each head and a generative prior has little left to add; where crowds are dense and heads shrink to a handful of pixels, the prior restores the structure the discriminative features discard.

Freezing is what keeps that prior worth having. Adapt the CLIP trunk in-domain and the diffusion gain shrinks but stays positive (+2.2 MAE under light LoRA); fine-tune the whole

backbone and the branch turns harmful, costing roughly 17.8 MAE on sparse B. We read this as partial substitution: a frozen generative prior stands in for most of what in-domain backbone training supplies. Once the trunk learns the target crowds it absorbs the regularities the prior had been supplying, and forcing the two to coexist only injects noise. Diffusion features behave the same way on correspondence, where a wide zero-shot margin narrows under supervision. So we deliberately freeze both backbones, and that decision is itself the result.

The method has clear limitations, which we state directly. On UCF-QNRF we trail the strongest trained counter, ChfL, and the leads on ShanghaiTech-A and ShanghaiTech-B hold only against the methods compared here. Most of the residual error lives in the extreme-density tail, the scenes above roughly a thousand people, where heads fall below the stride-8 grid. The diffusion branch shrinks that error but does not remove it: the stride-8 features and the 384-tile evaluation cap what one pass resolves in very-high-density scenes, so closing the rest will take higher-resolution fusion or adaptive tiling rather than a heavier neck, both left to future work. The density-axis reading covers two dense sets and one sparse set here, so a wider range of datasets is the right test for how far it generalizes. The frozen features also carry a generative content shift, which the minimal timestep $t=1$ holds down; distilling the branch into a noise-free, single-pass prior [31] would remove it.

Broader impact. Crowd counting sits close to surveillance, so privacy is a real concern here. DuoDiffCount outputs only an aggregate count, never identities or tracks, and a single forward pass needs no stored imagery. Responsible deployment should keep inference on device, retain no frames, and restrict who can query the counts. A count is a dual-use signal that can support crowd safety or feed monitoring, and that choice rests with the deployer.

One lesson here generalizes. The recipe we leaned on, a single-switch ablation, a capacity-matched control, a separation rule fixed before any run, and paired statistics stratified by density, is enough to strip out both the capacity confound and most of the run-to-run noise that makes fusion claims on small counting benchmarks hard to trust. None of it is specific to diffusion. Any such claim can be held to the same standard.

VI. CONCLUSION

DuoDiffCount counts crowds by fusing two frozen backbones, stable diffusion and CLIP-ConvNeXt, through a small trainable neck and density head in a single deterministic forward pass without sampling. With the backbones frozen, it records the lowest MAE and RMSE among the compared methods on both the sparse ShanghaiTech-B split (5.87) and dense ShanghaiTech-A, and stays competitive on UCF-QNRF.

The contribution the paper rests on is the attribution. A frozen diffusion prior measurably improves dense-scene counting, and a parameter-matched CLIP-only control shows the gain comes from the diffusion information, not the added capacity. The gain grows with crowd density and disappears on sparse scenes, so the B record is carried by the frozen base rather than the diffusion branch. It shrinks but persists once the discriminative

trunk is adapted in-domain, and under full fine-tuning, it reverses sign: a frozen generative prior largely substitutes for in-domain backbone training. Diffusion features show the same regime dependence on segmentation and correspondence; we observe the same in counting.

Two extensions are immediate. Distilling the diffusion branch into a lighter, noise-free prior would keep the dense-scene gain in an even leaner single pass. The extreme-density tail, scenes above roughly a thousand people, will need higher-resolution features or adaptive tiling rather than a richer neck.

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